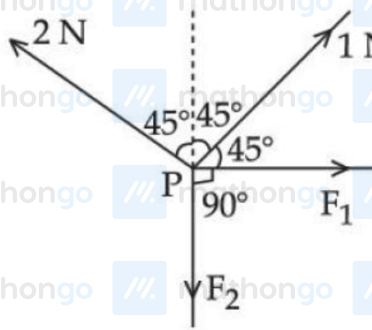


Q1 - 25 July - Shift 1

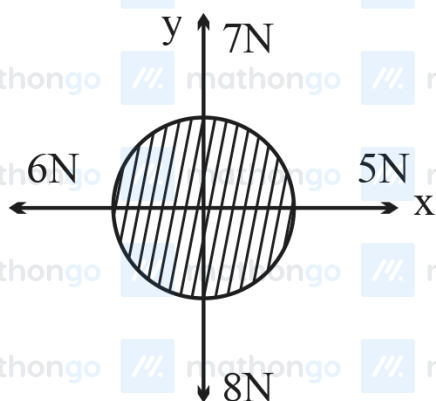
Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is $1 : x$ where $x = \underline{\hspace{2cm}}$.



Space for your notes:

Q2 - 25 July - Shift 2

For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero?



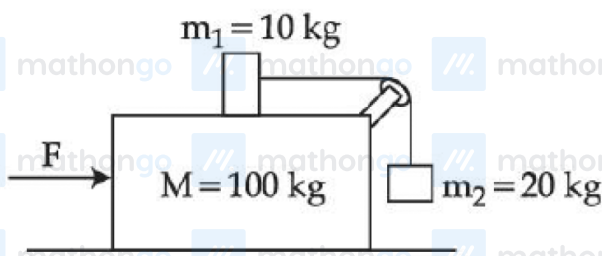
- (A) $\sqrt{2}$ N, 45° (B) $\sqrt{2}$ N, 135°
(C) $\frac{2}{\sqrt{3}}$ N, 30° (D) 2 N, 45°

Q3 - 26 July - Shift 1

Space for your notes:

Three masses $M = 100 \text{ kg}$, $m_1 = 10 \text{ kg}$ and $m_2 = 20 \text{ kg}$ are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2} . The value of F is :

(Take $g = 10 \text{ ms}^{-2}$)



- (A) 3360 N (B) 3380 N
(C) 3120 N (D) 3240 N

Q4 - 26 July - Shift 1

Space for your notes:

Questions

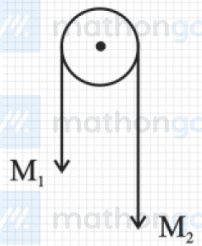
MathonGo

A monkey of mass 50kg climbs on a rope which can withstand the tension (T) of 350N. If monkey initially climbs down with an acceleration of 4m/s^2 and then climbs up with an acceleration of 5m/s^2 . Choose the correct option ($g = 10\text{m/s}^2$)

- (A) $T = 700\text{N}$ while climbing upward
- (B) $T = 350\text{N}$ while going downward
- (C) Rope will break while climbing upward
- (D) Rope will break while going downward

*Space for your notes:***Q5 - 26 July - Shift 2**

Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1 , the acceleration of the system is a_1 . When the mass M_2 is thrice that of M_1 , the acceleration of the system is a_2 . The ratio $\frac{a_1}{a_2}$ will be:

Space for your notes:

- (A) $\frac{1}{3}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{2}$
- (D) $\frac{1}{2}$

#MathBoleTohMathonGo

Q6 - 27 July - Shift 1

A bag is gently dropped on a conveyor belt moving at a speed of 2 m/s. The coefficient of friction between the conveyor belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion is : [Take $g = 10 \text{ m/s}^2$]

- (A) 2 m (B) 0.5 m
(C) 3.2 m (D) 0.8 ms

Space for your notes:

Q7 - 27 July - Shift 2

A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ . The magnitude of the contact force will be :

- (A) Mg
(B) $Mg \cos \theta$
(C) $\sqrt{Mg \sin \theta + Mg \cos \theta}$
(D) $Mg \sin \theta \sqrt{1 + \mu}$

Space for your notes:

Q8 - 27 July - Shift 2

Questions

MathonGo

A block 'A' takes 2 s to slide down a frictionless incline of 30° and length ' l ', kept inside a lift going up with uniform velocity ' v '. If the incline is changed to 45° , the time taken by the block, to slide down the incline, will be approximately:

- (A) 2.66 s (B) 0.83 s
(C) 1.68 s (D) 0.70 s

Space for your notes:

Q9 - 28 July - Shift 1

In two different experiments, an object of mass 5 kg moving with a speed of 25 ms^{-1} hits two different walls and comes to rest within

(i) 3 second, (ii) 5 seconds, respectively.

Choose the correct option out of the following :

- (A) Impulse and average force acting on the object will be same for both the cases.
(B) Impulse will be same for both the cases but the average force will be different.
(C) Average force will be same for both the cases but the impulse will be different.
(D) Average force and impulse will be different for both the cases.

Space for your notes:

Q10 - 28 July - Shift 1

#MathBoleTohMathonGo

Questions

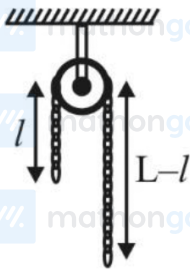
MathonGo

A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

- (A) 3 (B) 9 (C) 12 (D) 18

*Space for your notes:***Q11 - 28 July - Shift 2**

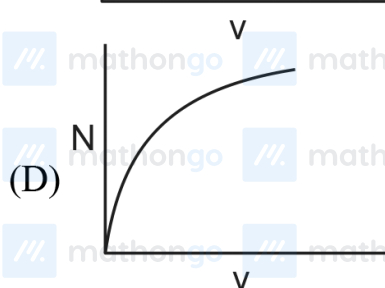
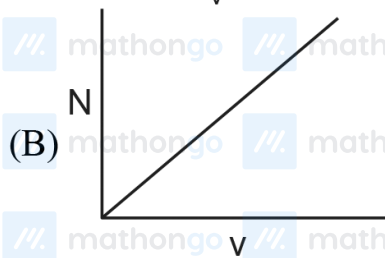
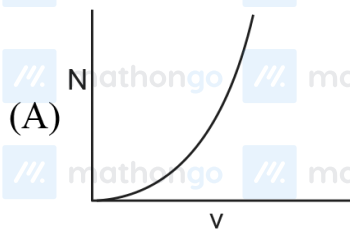
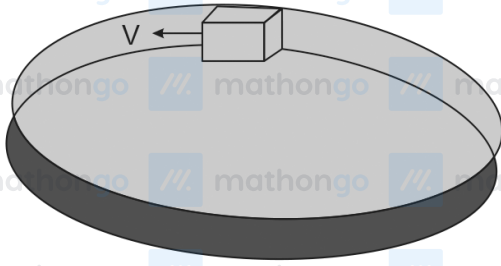
A uniform metal chain of mass m and length ' L ' passes over a massless and frictionless pulley. It is released from rest with a part of its length ' l ' is hanging on one side and rest of its length ' $L - l$ ' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is

Space for your notes:

- (A) 6 (B) 2 (C) 1.5 (D) 4

Q12 - 29 July - Shift 1

A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v . Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



Space for your notes:

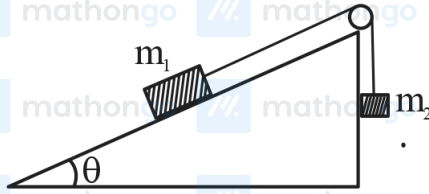
Q13 - 29 July - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

Two bodies of masses $m_1 = 5 \text{ kg}$ and $m_2 = 3 \text{ kg}$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be : [Take $g = 10 \text{ ms}^{-2}$]



- (A) 30 N (B) 40 N
(C) 50 N (D) 60 N

*Space for your notes:***Q14 - 29 July - Shift 2**

A tube of length 50 cm is filled completely with an incompressible liquid of mass 250 g and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F} \text{ rad s}^{-1}$. If F be the force exerted by the liquid at the other end then the value of x will be _____.

Space for your notes:

#MathBoleTohMathonGo

Answer Key

Q1 (3)

Q2 (A)

Q3 (A)

Q4 (C)

Q5 (B)

Q6 (B)

Q7 (A)

Q8 (C)

Q9 (B)

Q10 (B)

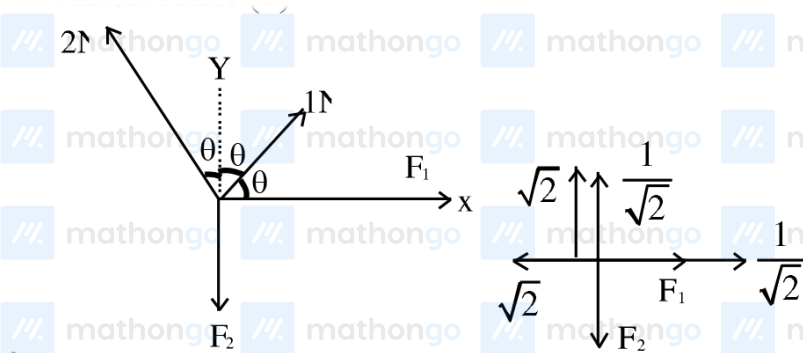
Q11 (D)

Q12 (A)

Q13 (B)

Q14 (4)

#MathBoleTohMathonGo

Q1 (3)

$$\theta = 45^\circ$$

Taking components along x & y

$$F_1 = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$F_2 = \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

$$F_1 : F_2 = 1 : 3$$

$$x = 3$$

Q2 (A)

Let addition force required is $= \vec{F}$

$$\vec{F} + 5\hat{i} - 6\hat{i} + 7\hat{j} - 8\hat{j} = 0$$

$$\vec{F} = \hat{i} + \hat{j}, |\vec{F}| = \sqrt{2}$$

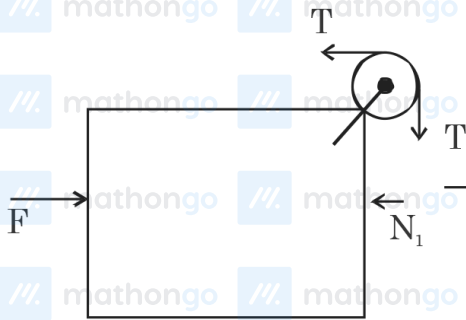
$$\text{Angle with x-axis: } \tan \theta = \frac{\text{y component}}{\text{x component}} = \frac{1}{1}$$

$$\theta = 45^\circ$$

Q3 (A)

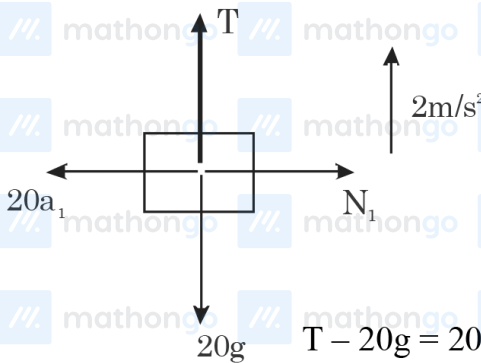
Let acceleration of 100 kg block = a_1

FBD of 100 kg block w.r.t ground



$$F - T - N_1 = 100 a_1 \dots\dots(i)$$

FBD of 20 kg block wrt 100kg

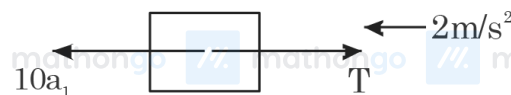


$$T - 20g = 20(2)$$

$$T = 240 \dots\dots(ii)$$

$$N_1 = 20a_1 \dots\dots(iii)$$

FBD of 10 kg block wrt 100 kg



$$10a_1 - 240 = 10(2)$$

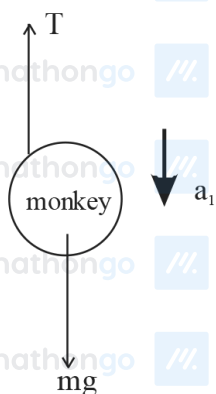
$$a_1 = 26 \text{ m/s}^2$$

$$F - 240 - 20(26) = 100 \times 26$$

$$\Rightarrow F = 3360 \text{ N}$$

Q4 (C)

F.B.D of monkey while moving downward

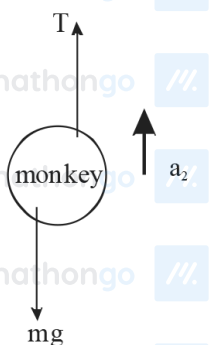


Using Newton's second law

$$mg - T = ma_1$$

$$\therefore 500 - T = 50 \times 4 \Rightarrow T = 300 \text{ N}$$

F.B.D of monkey while moving up



Using Newton's second law of motion

$$T - mg = ma_2$$

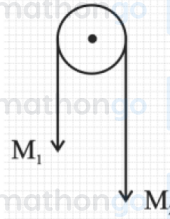
$$\Rightarrow T - 500 = 50 \times 5$$

$$\Rightarrow T = 750 \text{ N}$$

Breaking strength of string = 350 N

$\therefore$ String will break while monkey is moving upward

Q5 (B)



(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) $\frac{1}{2}$

Official Ans. by NTA (B)**Allen Ans. (B)**

Sol. $a = \frac{m_2 g - m_1 g}{m_1 + m_2}$

Case 1 $M_2 = 2m_1$

$$a_1 = \frac{2m_1 g - m_1 g}{3m_1}$$

$$a_1 = g/3$$

Case -2

$$M_2 = 3m_1$$

$$a_2 = \frac{3m_1 g - m_1 g}{4m_1}$$

$$a_2 = \frac{g}{2}$$

$$\frac{a_1}{a_2} = \frac{\frac{g}{3}}{\frac{g}{2}} = \frac{2}{3}$$

Q6 (B)

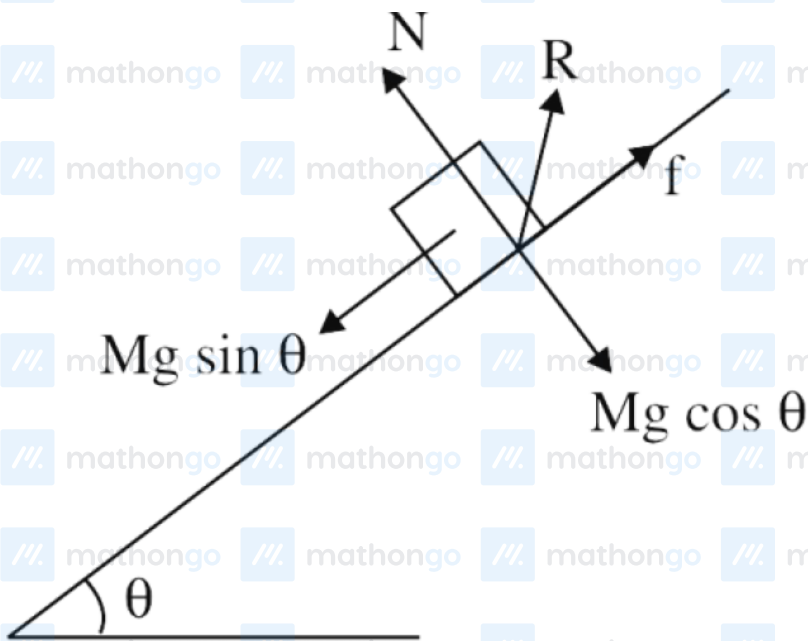
In frame of belt

$$a = \mu g = 4 \text{ m/s}^2, v = 2 \text{ m/s}, u = 0$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow s = 0.5 \text{ m}$$

Q7 (A)



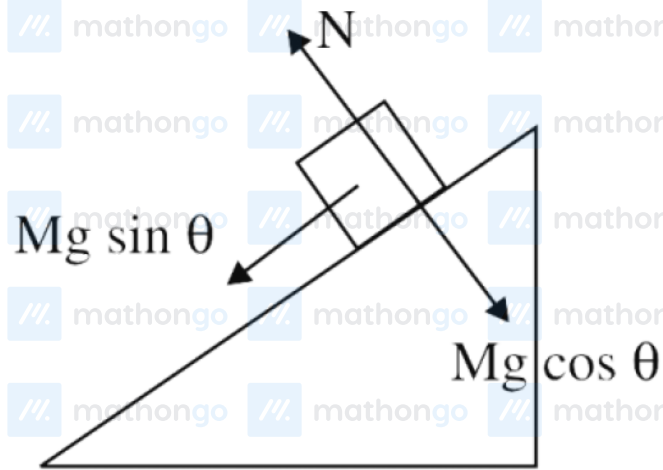
$$N = Mg \cos \theta$$

$$f = Mg \sin \theta$$

$$R = \sqrt{N^2 + f^2}$$

$$R = Mg$$

Q8 (C)



$$a = g \sin \theta$$

$$\ell = \frac{1}{2} g \sin 30^\circ (2)^2 \quad \dots\dots(1)$$

$$\ell = \frac{1}{2} g \sin 45^\circ t^2 \quad \dots\dots(2)$$

$$\left(\frac{1}{2}\right)(4) = \frac{1}{\sqrt{2}} t^2 \Rightarrow t = \sqrt{2\sqrt{2}} \approx 1.68$$

Q9 (B)

Impulse = change in momentum

$$I = \Delta P$$

$$F_{\text{avg}} = \frac{\Delta P}{\Delta t}$$

$$\Delta t_1 = 3 \quad \Delta t_2 = 5$$

$$\Delta P_1 = \Delta P_2$$

$$I_1 = I_2$$

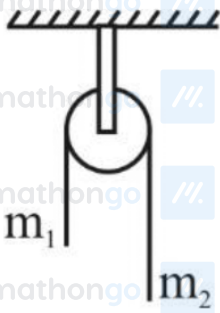
F_{avg} in case (i) is more than (ii)

Q10 (B)

$$F = \frac{dm}{dt} v$$

$$= \frac{10g}{5s} \left(4.5 \frac{\text{cm}}{s} \right) = 9 \frac{g\text{cm}}{s^2} = 9 \text{ dyne}$$

Q11 (D)



$$a = \frac{(m_2 - m_1)}{(m_2 + m_1)} g$$

$$\frac{g}{2} = \frac{(\lambda(L - \ell) - \lambda\ell)g}{\lambda L} \Rightarrow L = \frac{L}{4} = \frac{L}{x}$$

$$x = 4$$

Q12 (A)

From this, (A)

$$N = \frac{mv^2}{r}$$

Curve is parabola

$$Y = kx^2$$

Q13 (B)

For equilibrium $m_2 g = m_1 g \sin \theta$

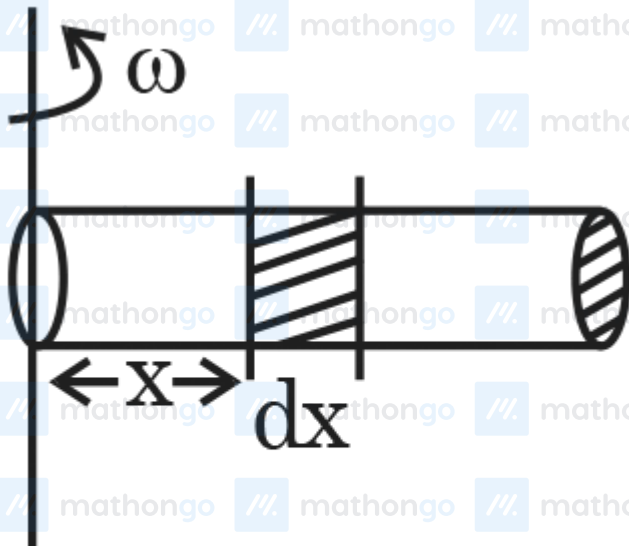
$$\sin \theta = \frac{m_2}{m_1} = \frac{3}{5}$$

$$\cos \theta = \frac{4}{5}$$

Normal force on $m_1 = 5g \cos \theta$

$$= 5 \times 10 \times \frac{4}{5} = 40 \text{ N}$$

Q14 (4)



$$F = \int (dm) \omega^2 x$$

#MathBoleTohMathonGo

$$= \int_0^L \left(\frac{m}{L} dx \right) \omega^2 x$$

$$= \frac{m}{L} \omega^2 \frac{L^2}{2}$$

$$= \frac{m \omega^2 L}{2}$$

$$\omega = \sqrt{\frac{2}{mL}} \sqrt{F}$$

$$= \sqrt{\frac{2}{0.25 \times 0.5}} \sqrt{F}$$

$$= \sqrt{16} \sqrt{F}$$

$$= 4 \sqrt{F}$$

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